The Turing Center at the University of Wash.

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Turing Center

UNIVERSITY OF WASHINGTON

Established May 2005

Investigating problems at the crossroads of natural language processing, machine learning, Web search, and the Semantic Web.

http://turing.cs.washington.edu

TURING CENTER MISSION

- Now is the time to make world-wide all-to-all communication feasible.
- Our Mission is to enhance the philosophy, science, and technology of pan-lingual communication and collaboration among human and artificial agents.

INITIAL RESEARCH FOCI

- Division of labor between machine and human learning.
- Design multilingual human-machine multiagent systems.
- Human-machine collaboration in ambiguity control and translation.
- Pan-lingualization of the Semantic Web.
- Leverage natural language in meaning representation.

STRUCTURE

- Launch date: May 1, 2005
- Location, Computer Science & Engineering, UW
- Director: Oren Etzioni
- Advisory committee: Emily Bender, Oren Etzioni, David Notkin, Jonathan Pool (chair)
- Funding $5 million (gift from Utilika Foundation, grants)

AMBIGUITY CONTROL

- How much can disambiguation at the source improve MT?
- Under what circumstances will authors feel it worth their while to disambiguate?
- How can we make it easy for them to do so (given point-to-point or broadcast MT)?

INTEREST IN DELPH-IN

- Tasked with extending to under resourced as well as well-resourced languages.
- Controlled languages/disambiguation at the source work better with precision grammars.
- LKB +Grammar Matrix + LOGON transfer machinery seem like an interesting avenue to explore

TOWARDS PLMT

- Rapid morphological analyzer development
- Extend the phenomena handled in Matrix customization
- Rapid lexical acquisition for Matrix seedlings
- ML for translation lexicons
- Autogeneration from Matrix of transfer rules for grammatical concepts

CONTROLLED LANGUAGE

- Use Matrix customization information and bilingual translation dictionary to ask for clarification where needed.
- Use discriminants to present author with (understandable!) choices about which meaning is intended.
- ... lead users to expect less than complete expressivity.